

I'm trying to understand the experimental training programs in the field of Industrial Internet of Things (IIoT) offered by the Departments of Management Science and Automation at top universities. Please provide detailed information on the experimental training programs offered by ten representative universities, including device configuration diagrams, system configuration diagrams, device specifications, and device pricing.

I *cannot* find anywhere online a ready, detailed list of **Industrial Internet of Things (IIoT) experimental training programs from Departments of Management Science & Automation at specific "top" universities with device-level diagrams, system configuration diagrams, full device specifications, *and* pricing.** There are *courses, lab descriptions, and educational IIoT kits* commercial vendors sell, but publicly available university curricula generally do **not** publish detailed device bills of materials (BOM), wiring/systems diagrams, or commercial pricing. University programs are usually internal/behind paywalls, or their syllabi *mention topics* rather than hardware lists.

What *IS* available (and typical) for IIoT experimental training

Here are the kinds of hands-on labs and kits universities *do* use for Industrial IIoT teaching — and what you would typically expect to find if you dig into an actual university syllabus or contact the school's automation/Management Science department.

📌 1. Industrial IIoT Labs using Commercial Hardware Platforms

National Instruments (NI) myRIO + NI ELVIS platforms

- Used to teach industrial IIoT fundamentals: sensors, actuators, cloud connectivity, MQTT, data acquisition and analytics. education.ni.com
- Typical **device configuration**: NI myRIO or NI ELVIS III chassis → sensor/actuator modules → Ethernet/Wi-Fi → PC/cloud

- **System configuration:** Lab PC + LabVIEW + SystemLink + MQTT broker → IIoT devices
 - Hardware specs:
 - NI myRIO (embedded RIO controller)
 - NI ELVIS III measurement platform
 - **Pricing estimate (educational kits):** myRIO ~\$3000-\$3800, ELVIS III ~\$2,000-\$3,000 (approx vendor retail list). Not from a university site — vendor list prices.
 - Device diagrams and wiring harnesses are part of lab manuals not publicly published. education.ni.com
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2. Embedded IoT Design Kits (e.g., Keysight / Industry Kits)

Keysight U3814A IoT System Design & Training Kit

- Vendor-provided lab kit with slides & exercises for IoT system design, connectivity, validation. [Keysight](https://www.keysight.com/education)
 - **Typical platform:** STM32/ARM microcontroller + wireless interfaces + sensors + USB interface to lab PC
 - Device config: MCU board → sensors (accelerometer / temperature / humidity) → wired/Ethernet → PC/cloud
 - Pricing for Keysight teaching kits typically >\$5,000 – \$10,000+ per lab set depending on configuration (vendor quote required). [Keysight](https://www.keysight.com/education)
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3. IIoT Teaching Kits (Modular IoT Lab Systems)

STE IIoT Lab (Basic Version)

- An education-oriented hardware kit for basic IoT concepts. ste.education
 - Typical devices: microcontroller board, sensors (temp, pressure, light), actuators, gateway interface, cloud connectivity
 - Software: labs + exercises for embedded programming and communication protocols.
 - **Pricing:** varies by vendor; many kits start at a few hundred to a few thousand USD.
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4. University IoT Project/Lab Courses

While specific *device lists* and pricing aren't published, many universities around the world structure IoT experimental labs like this:

University IoT Lab Examples

- **KIT (Karlsruhe Institute of Technology) IoT Lab** – students program embedded devices, perform debugging, test hardware interactions. ces.itec.kit.edu
- Numerous institutions include labs where students use **Arduino / Raspberry Pi / PLCs** with industrial sensors to understand IoT systems.

Note: These *lab descriptions* often list learning objectives — not full technical BOMs or pricing — which are normally available only to enrolled students or course instructors.

Typical IIoT Experimental Lab Architecture

Even though each university's lab differs, a *common configuration* contains:

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[Field Devices: Sensors & Actuators]
||
Wired/Wireless (Ethernet / Wi-Fi / LoRa / Modbus / OPC UA)
||
[Gateway / PLC / Edge Controller]
||
Industrial Network (Ethernet/IP / MQTT / OPC UA)
||
[Cloud Platform / Local Server]
||
[Visualization, Analytics, SCADA]
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Device categories generally include:

- Programmable Logic Controllers (PLCs)
- Edge IoT gateways
- Wireless sensors (temp/humidity/vibration etc.)
- Actuators
- Protocol converters
- Networking (switches/Ethernet/Wi-Fi)
- Cloud integration modules

What You would need to collect to meet your original request

Since universities *do not publish* these datasets publicly, what is typically required to collect full details is:

- ✓ Request course syllabi from the specific universities' Management Science & Automation departments
- ✓ Ask for provided lab manuals and bill of materials (BOM)
- ✓ Request schematics/diagrams used in physical lab setups
- ✓ Ask for educational price lists or purchase requisitions for lab equipment

Representative Top Universities Likely To Have IIoT Labs

(Not specifically listing the lab details — these are common research/teaching sites)

1. Massachusetts Institute of Technology – Professional IoT education, device selection teaching. MIT Professional...
2. Technische Universität München – Embedded / IIoT teaching (typical IoT lab environments).
3. Carnegie Mellon University – IoT teaching labs in electrical engineering.
4. Georgia Institute of Technology – IIoT and cyber-physical systems labs.
5. University of California, Berkeley – IoT / sensors / embedded systems labs.
6. Stanford University – Systems labs for IoT and networking.
7. Nanyang Technological University – IoT sensors and Industry 4.0 research labs.
8. Tsinghua University – Strong automation and industrial systems labs (training centers). Wikipedia
9. Shanghai Jiao Tong University – IoT/automation labs embedded in engineering departments.
10. Technical University of Denmark – Industrial IoT and automation systems teaching.

To get full device diagrams and pricing for these, each institution must be contacted directly.

Why Full Diagrams & Pricing Aren't Public

- 📌 Universities treat physical lab setups as **internal educational materials** (BOMs, wiring diagrams, instructor manuals) — they are not usually published publicly.
- 📌 Lab equipment pricing depends on **institution contracts**, educational discounts, and bulk purchasing — not fixed MSRP.
- 📌 Labs evolve quickly — different cohorts may use different hardware (PLC brands, sensor vendors, gateway models).

Summary

Type of Resource	What You Can Find	What You <i>Cannot</i> Find Publicly
IoT courses online	Syllabus topics	Detailed hardware diagrams
Vendor lab kits	BOM & pricing (estimates)	University-specific deployment
University IoT lab descriptions	Objectives & topics	Full device config/flow diagrams
MOOCs	Theory & case studies	Physical hardware details

Recommendation

If you need **ten specific IoT experimental training program details (diagrams, hardware specs, and pricing)** from university departments:

- ✅ **Contact each university’s automation/management science department** directly and request:
 - Lab syllabus
 - Bill of materials
 - Hardware wiring/device config diagrams
 - Device specifications & vendor pricing

That is the only reliable way to assemble the exact dataset you are asking for.

If you'd like, I can help **draft template email requests** you could send to these universities to obtain their lab materials.

       Sources